A PRACTICAL DISSECTION GUIDE

ABDOMINAL BLOOD VESSELS

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1. Orientation

- Lay the specimen ventral side uppermost and pin it securely through the feet of the anterior limbs stretched antero-laterally and posterior limbs postero-laterally.

2. Skinning

- Pinch and make a mid-ventral incision through the skin from the anus in the pubic region up to the mandibles/lower jaw.
- Then make transverse lateral cuts from the throat to the base of each forelimb and from the pubis to the hind limbs.
- The Skin is loosened from the body wall, and its flaps are laterally pinned back.

3. Body Wall Opening

- Elevate the abdominal wall and make a small incision through the linea alba just anterior to the pelvic girdle. Extend the median incision anteriorly to the xiphoid.
- Make lateral incisions postero-laterally near the hindlimbs and anterior-laterally along the costal margin. Pin the muscular flaps laterally to expose the structures in the abdominal cavity.

4. Displaying Abdominal Blood Vessels

- Lay the stomach and intestines over to the right. Clear fat carefully around the coeliac artery and anterior mesenteric artery to display them.
- Identify and ligature the hepatic portal vein, coeliac artery and anterior mesenteric artery distal to the ligatured points.

4. Displaying Abdominal Blood Vessels

- Make a transverse cut through the upper end of the esophagus and the distal end of the rectum after ligaturing the coeliac and related vessels.
- Gently tear the mesentery and remove the alimentary canal carefully.
- Clear remaining fat and connective tissue from the dorsal wall to expose the full arterial and venous tree, including the following vessels:
- Abdominal/dorsal aorta, Posterior vena cava, Coeliac artery, Anterior mesenteric artery, Posterior mesenteric artery, Renal arteries and veins, Hepatic portal vein, Right and left gastric arteries and veins, Common iliac arteries and veins, External and internal iliac arteries and veins

Summary of displacements to display Abdominal vessels

Vessel Exposed	Part Cut / Ligatured	Part Displaced + Direction
Abdominal aorta	Alimentary canal after the ligatures	Fat and connective tissue — Cleared from dorsal wall
Posterior vena cava	Alimentary canal after the ligatures	Fat and connective tissue — Cleared from dorsal wall
Coeliac artery	Ligature the vessel	Stomach and liver — To the right / anteriorly or removed
Anterior mesenteric artery	Ligature the vessel	Small intestine loops — Laterally to the right or removed
Posterior mesenteric artery	Ligatured, rectum cut out	Descending colon — Laterally to the left or removed
Renal arteries and veins	Perirenal fat (cleared)	Kidneys — Slightly laterally

Summary of displacements to display Abdominal vessels cont'd

Hepatic portal vein	Ligature near portal area	Duodenum / gut — Forced through cut / to the right
Gastric arteries and veins	Related mesentery (cleared)	Stomach — To the right
Common iliac arteries and veins	Alimentary canal (after ligatures)	Pelvic fat — Cleared
External and internal iliacs	Alimentary canal (after ligatures)	Hind limb tissues — Slightly laterally

Abdominal vessels displays

ABDOMINAL VASCULAR TREE

PREPARE	LIGATURE	REMOVE AND DISPLAY
Open abdomen once Lay gut to the right Clear fat around vessels	Identify and ligature Hepatic portal vein Coeliac artery Anterior mesenteric artery	Make cuts at the oesophagus and rectum Gently tear the mesentery Remove alimentary canal Clear dorsal wall fat Expose full vessel tree